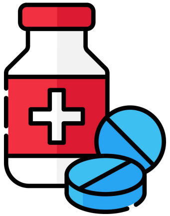
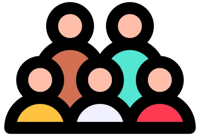
Attrition Insights Briefing

- Mati Borlak
- Brenna Horowitz
- Aviv Goldshide
- Danila Stepanov

Dataset Overview



- 1470 employees at pharmaceutical/biotech/medical company
- 3 categories of features:



Demographics & Tenure

- Age
- Gender
- Education
- Years at company
- etc.

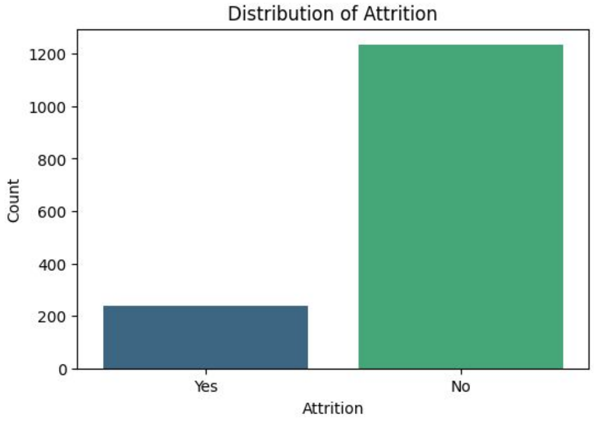
Job-Related Factors & Perceptions

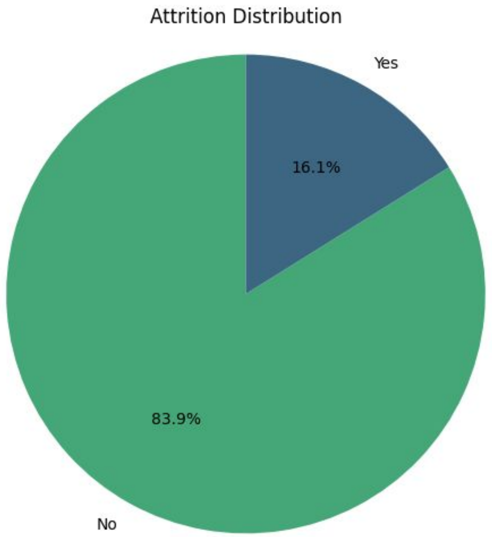
- Job Satisfaction
- Work life balance
- Performance rating
- etc.

Compensation & Benefits

- Monthly Income
- Stock options
- % salary hike
- etc.

Target Variable





- Attrition often refers to departures that are not replaced, or sometimes only voluntary resignations.
- The average attrition rate in the pharmaceutical industry is notably high.
- Recent data indicates that the voluntary turnover rate in this sector is approximately 13%.
- We have no historical context.

Insight 1 - Money & Position

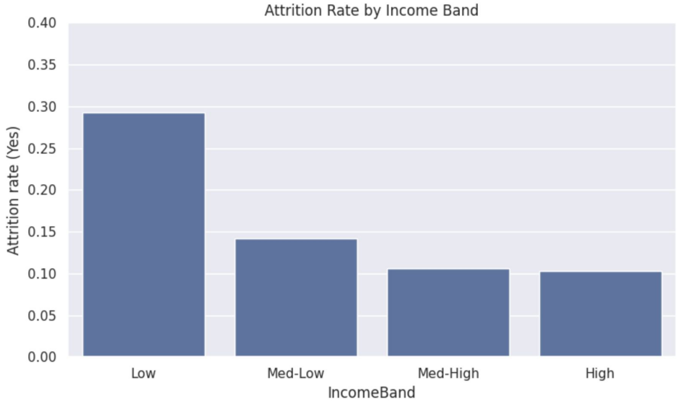
Lower-income employees consistently show the highest attrition within every job level, revealing that compensation relative to job expectations—not seniority alone—is a major driver of turnover.

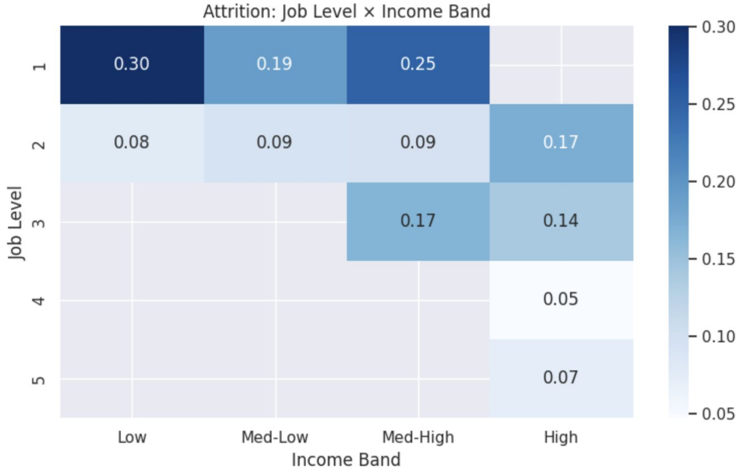
- Lowest income band employees within each Job Level have the highest attrition (2–3× higher than higher-paid peers).
- Attrition decreases as income increases, even when Job Level is constant.
- High-income employees at higher Job Levels show the lowest overall attrition.

Employees compare pay to peers at the same job level.

- Lower-income employees often feel:
  - underpaid compared to peers
  - limited growth opportunities
  - more attracted to external offers
  - more financially stressed
- Higher-income employees tend to feel:
  - valued and secure
  - financially stable
  - more committed to long-term benefits

Adding Department shows Sales employees in the lowest income bands churn at the highest rates, while R&D and HR show a smoother decline from low to high income bands.



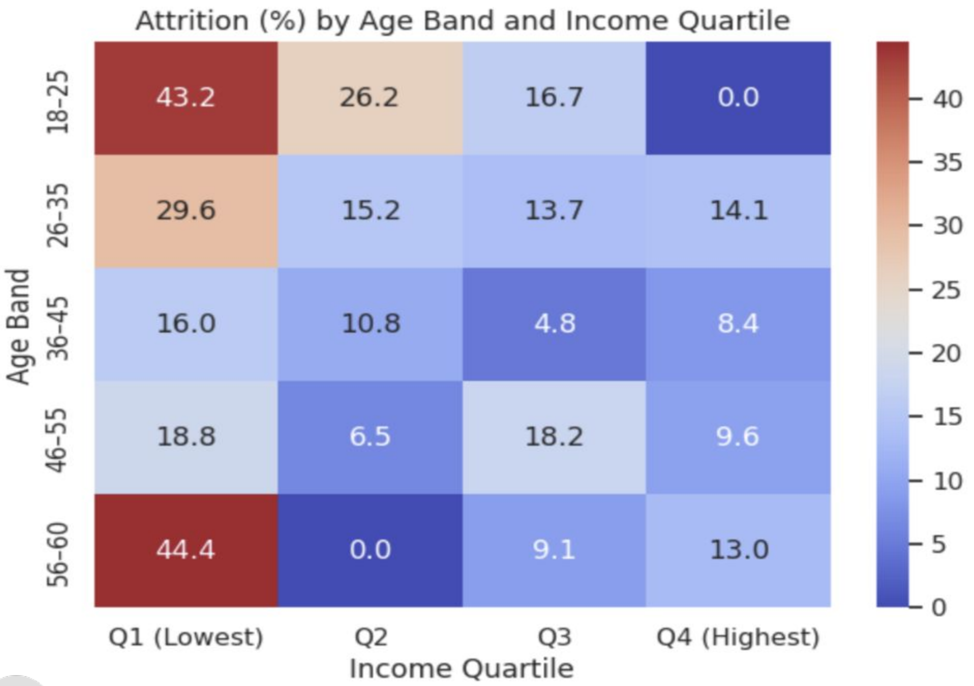


Insight 2 - Money & Age

High attrition among younger workers is concentrated in lower income groups, or if age effects persist regardless of pay.

- Young, low-income employees have the highest attrition:
  - Ages 18–25 in Q1 show 43% attrition.
  - Ages 26–35 in Q1 show 30% attrition.
- Higher income significantly reduces attrition for younger workers:
  - For ages 18–25, attrition drops from 43% (Q1) to 0% (Q4).
  - For ages 26–35, it drops from 30% (Q1) to 14% (Q4).
- Mid-career employees (36–45) are stable across all income levels:
  - Attrition ranges only from 5% to 16%.
- Older employees (56–60) mirror the young group:
  - Very high attrition in Q1 (44%), which is lower in higher income groups.

Income impact is greatest at career start and end. Low-income young and older employees are most likely to leave, while mid-career employees remain stable regardless of pay.



Insight 3 - Overwork

Employees with overtime and low work-life balance (WLB scores 1–2) exhibit the highest company attrition, significantly exceeding all other groups (e.g., those with overtime but high WLB scores 3–4, or non-overtime employees with high WLB, who have the lowest attrition). Overtime indicates high job intensity and potential burnout, while low WLB signals lack of control and dissatisfaction. This combination leaves employees feeling:

- overwhelmed
- time-starved
- misaligned with company culture

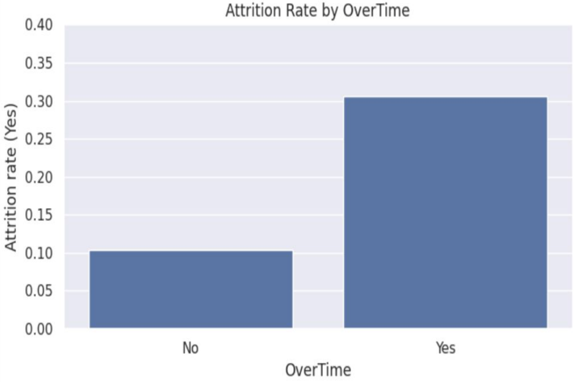
Sales and HR departments with low WLB and overtime show extreme attrition peaks (35%–40%); R&D follows a similar, slightly lower pattern.

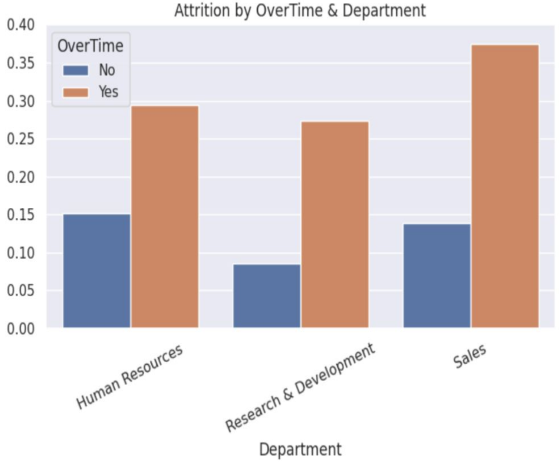
OverTime × Work Life Balance × Attrition:

Attrition	OverTime	WLB_Band	No	Yes
0	No	1	0.741379	0.258621
1	No	2	0.891667	0.108333
2	No	3	0.915493	0.084507
3	No	4	0.871795	0.128205
4	Yes	1	0.545455	0.454545
5	Yes	2	0.692308	0.307692
6	Yes	3	0.712598	0.287402
7	Yes	4	0.666667	0.333333

OverTime × Department × Attrition:

Attrition	Department	OverTime	No	Yes
0	Human Resources	No	0.847826	0.152174
1	Human Resources	Yes	0.705882	0.294118
2	Research & Development	No	0.914493	0.085507
3	Research & Development	Yes	0.726937	0.273063
4	Sales	No	0.861635	0.138365
5	Sales	Yes	0.625000	0.375000





Insight 4 - Satisfaction

Job satisfaction critically affects attrition, with the impact varying by department, especially in high-pressure Sales and R&D. Satisfaction level 1 employees have the highest attrition across all departments. Attrition drops universally as satisfaction rises from 1 → 4, with the steepest decline in Sales and R&D. Satisfaction level 4 employees have some of the lowest attrition company-wide.

Different departments have distinct stress profiles:

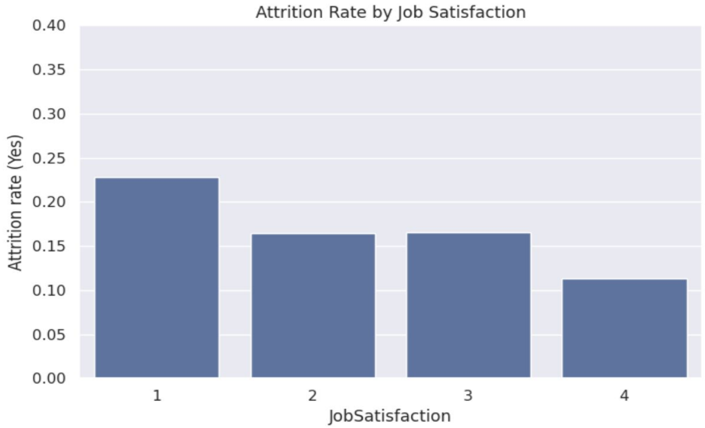
- Sales: High pressure, quotas, competitive targets.
- R&D: High cognitive load, deadlines, performance expectations.
- HR: Emotional labor, conflict management.

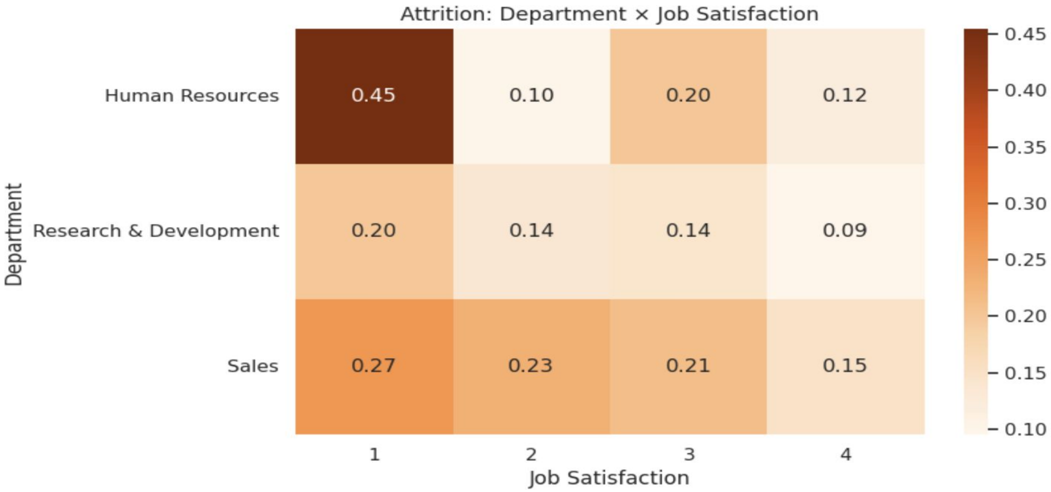
Low satisfaction in these environments signals deep issues:

- Burnout
- Unclear growth paths
- Poor manager relationships
- Mismatched expectations

Sales employees with satisfaction level 1 experience the highest attrition rates overall.

R&D employees with Satisfaction level 4 show extremely low attrition, proving satisfaction stabilizes high-pressure roles.





Insight 5 - Education Fit

Attrition is low when the job matches the education field.

- Example: Life Sciences graduates in scientific roles have very low attrition (3–14%).

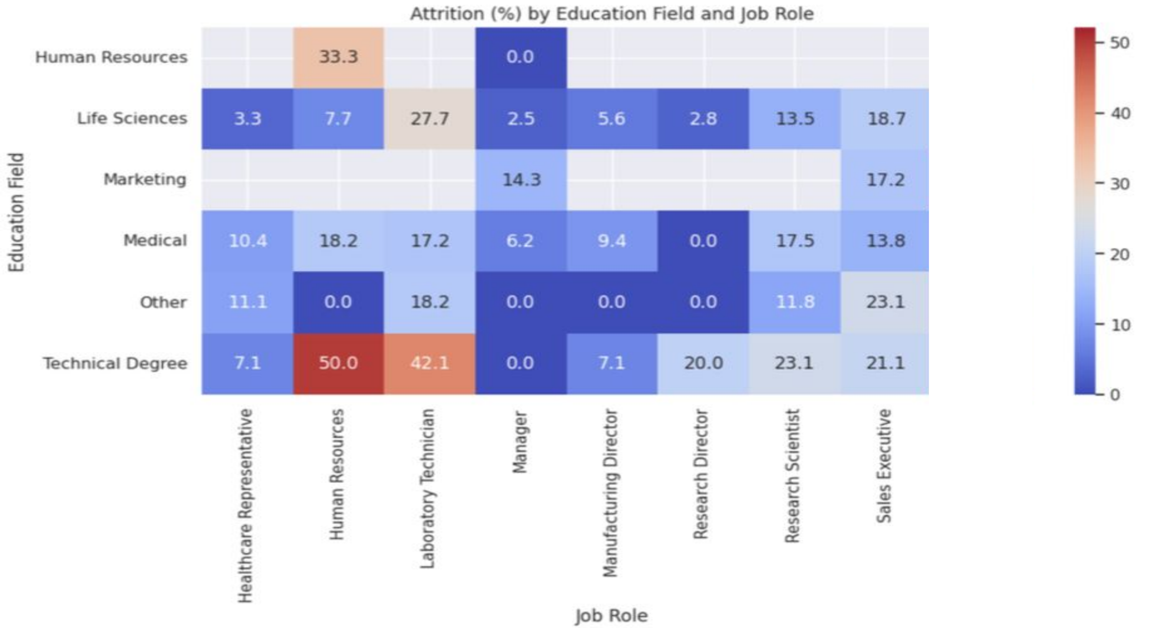
Attrition is much higher when employees work outside their field.

- Technical Degree graduates show high attrition in HR (50%), and Lab Tech (42%).

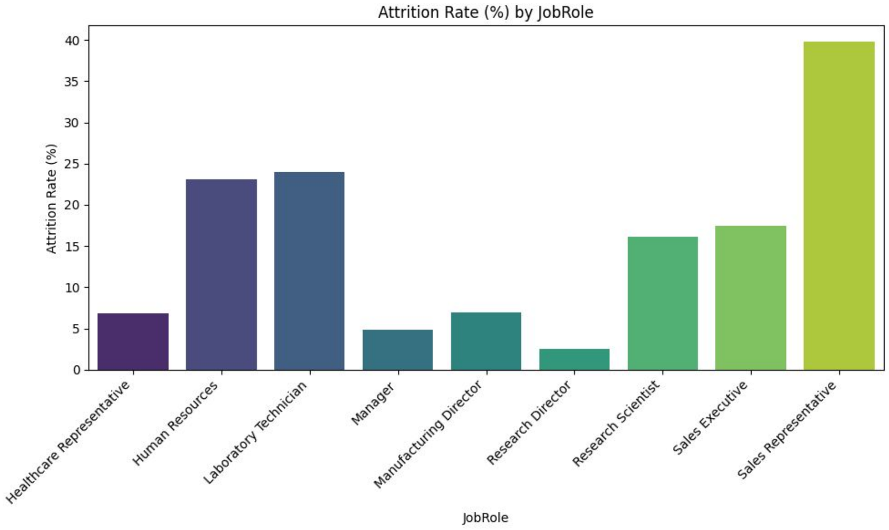
Sales roles have the highest turnover across all dimensions, they are a clear exception to be investigated.

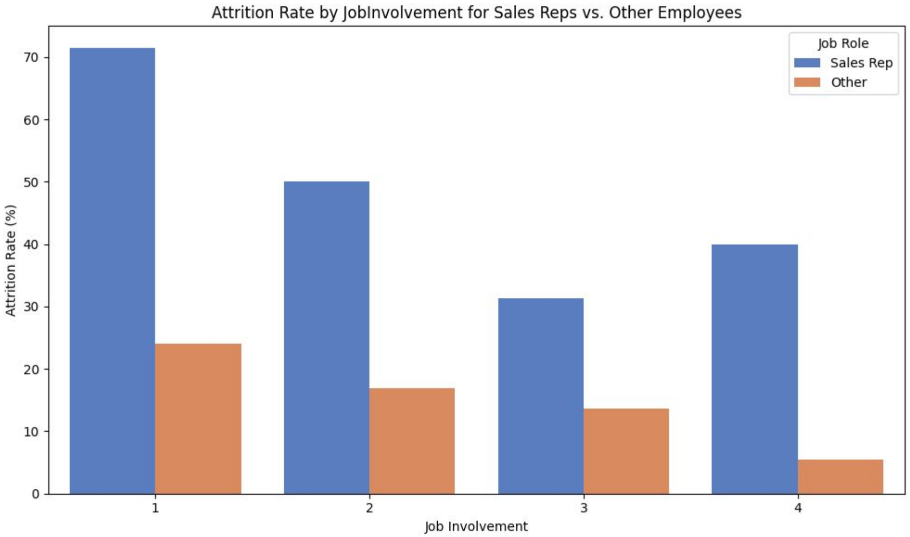
Scientific and managerial roles are the most stable.

Employees stay longer when their job aligns with their education; turnover increases sharply in non-matching roles.



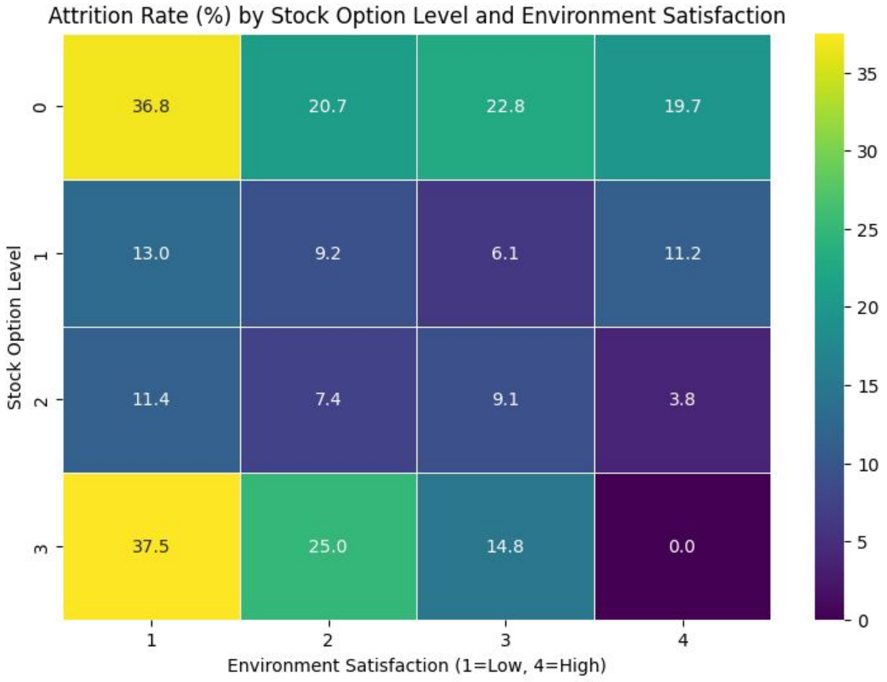
Insight 6 - Sales Reps

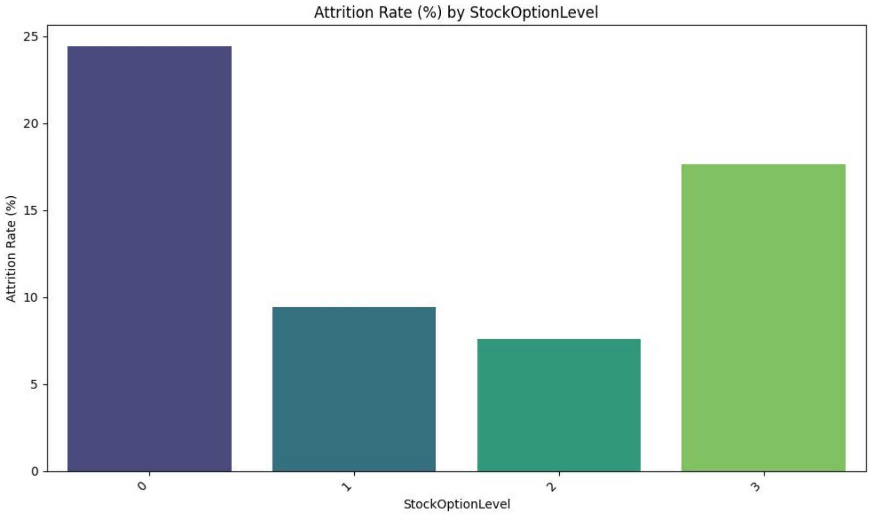




- Pharmaceutical sales representatives experience higher turnover rates. Statistics show that about 35% of sales reps leave their positions, with 44% departing within the first 1-2 years.
- The highest attrition rate is observed in sales reps with very low involvement, exceeding 70%. This highlights a critical area for intervention within the sales department.
- In sales, performance is often directly tied to compensation (commissions, etc.). Low job involvement can translate to less effort, poorer performance, and consequently, lower earnings.

Insight 7 - Stock Options & Environment





- We expect employees with the best stock options to have lowest attrition rate.
- This could be countered by non-financial reasons, new opportunities, vesting reasons, or projected decline in company stock.
- High stock options only become a significant retention factor when the environment is also satisfactory.
- Poor management, poor culture, toxic.

Conclusions

- Attrition is not random; it clusters around predictable factors.
- Low-income employees, especially early-career and sales reps, consistently show the highest turnover across multiple insights.
- Employees working outside their education field or in mismatched roles experience sharp increases in attrition.
- Workload pressure + low work-life balance is a strong predictor of attrition, acting as a multiplier of risk.
- Job satisfaction plays a stabilizing role. High satisfaction dramatically reduces attrition even in high-stress departments.
- Targeted improvements in compensation fairness, job-role alignment, workload distribution, and manager support would significantly improve retention.